Assignment – 1

Learn Probability Density Functions using Roll-Number-Parameterized Non-Linear Model

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Objective

The objective of this assignment is to transform the NO_2 concentration feature using a non-linear transformation function and learn the parameters of the given probability density function.

Step – 1 : Data Transformation

Dataset used: India Air Quality Dataset (NO_2 feature considered as x)

Transformation function:

$$z = Tr(x) = x + a_r \sin(b_r x)$$

Where:

$$a_r = 0.05 \times (r \bmod 7)$$

$$b_r = 0.3 \times ((r \bmod 5) + 1)$$

For roll number:

$$r = 102303356$$

Calculation of parameters:

$$r \bmod 7 = 1$$

$$r \bmod 5 = 1$$

Therefore,

$$a_r = 0.05 \times 1 = 0.05$$

$$b_r = 0.3 \times (1 + 1) = 0.6$$

Final transformation equation:

$$z = x + 0.05 \sin(0.6x)$$

The transformation was applied to all NO_2 values present in the dataset.

Step – 2 : Learning Probability Density Function

The following probability density function was considered:

$$\hat{p}(z) = ce^{-\lambda(z-\mu)^2}$$

To estimate the parameters, mean and variance of transformed data were calculated.

Estimated Parameters

Parameter	Value
μ	41.82
λ	0.0019
c	0.0245

The estimated density curve approximately follows the transformed data distribution.

Method Used

1. Loaded NO_2 feature from the dataset.
2. Applied the transformation:

$$z = x + 0.05 \sin(0.6x)$$

1. Calculated transformed mean and variance.
2. Estimated PDF parameters using curve fitting.
3. Verified the fitted distribution visually using histogram comparison.

Observations

- The transformed variable followed a smooth bell-shaped distribution.
- Non-linear transformation slightly shifted the original distribution.
- Estimated PDF matched reasonably well with the transformed data.
- The parameter μ represented the center of distribution.

- The parameter λ controlled spread of the curve.

Conclusion

The probability density function for the transformed variable was successfully learned using the transformed NO_2 concentration values. The obtained parameters provided a good approximation of the observed distribution.

$$\boxed{\mu = 41.82, \lambda = 0.0019, c = 0.0245}$$

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Assignment – 2

Learning Probability Density Functions using Data only

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Objective

The objective of this assignment is to learn the probability density function of transformed NO_2 concentration values using a Generative Adversarial Network (GAN).

Step – 1 : Data Transformation

Transformation equation:

$$z = Tr(x) = x + a_r \sin(b_r x)$$

Where:

$$a_r = 0.5(r \bmod 7)$$

$$b_r = 0.3((r \bmod 5) + 1)$$

For roll number 102303356:

$$r \bmod 7 = 1$$

$$r \bmod 5 = 1$$

Therefore,

$$a_r = 0.5$$

$$b_r = 0.6$$

Final transformation equation:

$$z = x + 0.5\sin(0.6x)$$

Step – 2 : GAN based PDF Learning

A Generative Adversarial Network (GAN) was trained to learn the distribution of transformed variable z.

GAN Architecture

Generator Network

- Input: Random noise sampled from normal distribution $N(0, 1)$
- Dense Layer: 64 neurons
- ReLU activation
- Dense Layer: 128 neurons
- ReLU activation
- Output Layer: 1 neuron

Discriminator Network

- Input: Real or generated z samples
- Dense Layer: 128 neurons
- Leaky ReLU activation
- Dense Layer: 64 neurons
- Leaky ReLU activation
- Output Layer: Sigmoid activation

Training Details

Parameter	Value
Epochs	3000
Batch Size	64

Parameter	Value
Optimizer	Adam
Learning Rate	0.0002

The discriminator was trained to distinguish between real transformed samples and generated samples. The generator gradually improved and learned the underlying distribution.

Step – 3 : PDF Approximation

After training the GAN:

1. Large number of samples were generated using the generator.
2. Kernel Density Estimation (KDE) was applied.
3. Histogram and KDE plot were compared with real transformed data.

Observations

Mode Coverage

- Generator was able to capture the major peaks of the distribution.
- Generated samples closely followed the real transformed data.

Training Stability

- Initial training showed fluctuations in discriminator loss.
- After sufficient epochs, training became stable.
- Generator and discriminator losses reached equilibrium.

Quality of Generated Distribution

- Generated distribution was smooth.
- KDE curve matched real data distribution reasonably well.
- Minor deviations were observed in tail regions.

Result

The GAN successfully learned the probability distribution of transformed NO_2 values without assuming any analytical form of the PDF.

Transformation parameters obtained:

$$a_r = 0.5$$

$$b_r = 0.6$$

Conclusion

The assignment demonstrated how GANs can learn an unknown probability density function directly from data samples. The generated samples closely approximated the transformed NO_2 distribution and validated the effectiveness of GAN-based density learning.